4	**(a)**	**(i)**	$x = 18.0$ cm	**A1**
		(ii)	$V_A + V_B = 0$ $\dfrac{3.6\times10^{-9}}{4\pi\varepsilon_0(0.180)} + \dfrac{q_B}{4\pi\varepsilon_0(0.120)} = 0$ $q_B = -2.4 \times 10^{-9}$ C (use of $V_A = V_B$ giving 2.4×10^{-9} C scores one mark)	**C1** **C1** **A1**
	(b)		By conservation of energy, Total energy at $x = 5.0$ cm = total energy at $x = 27.0$ cm $e(560) + KE_A = e(-600) + KE_B$ (= 0 J) $KE_A = 1.9 \times 10^{-16}$ J $\frac{1}{2} mv^2 = 1.9 \times 10^{-16}$ J Speed of electron = 2.02×10^7 m s^{-1}	**C1** **C1** **A1**
	(c)		The gradient gives the magnitude of the net E-field strength. Since force experienced equals to charge multiply with net E-field The largest force experienced is at $x = 27.0$ cm since the gradient is the largest.	**B1** **B1** **B1**

5	**(a)**	**(i)**	Total charge = $nAL\,e$	**A1**